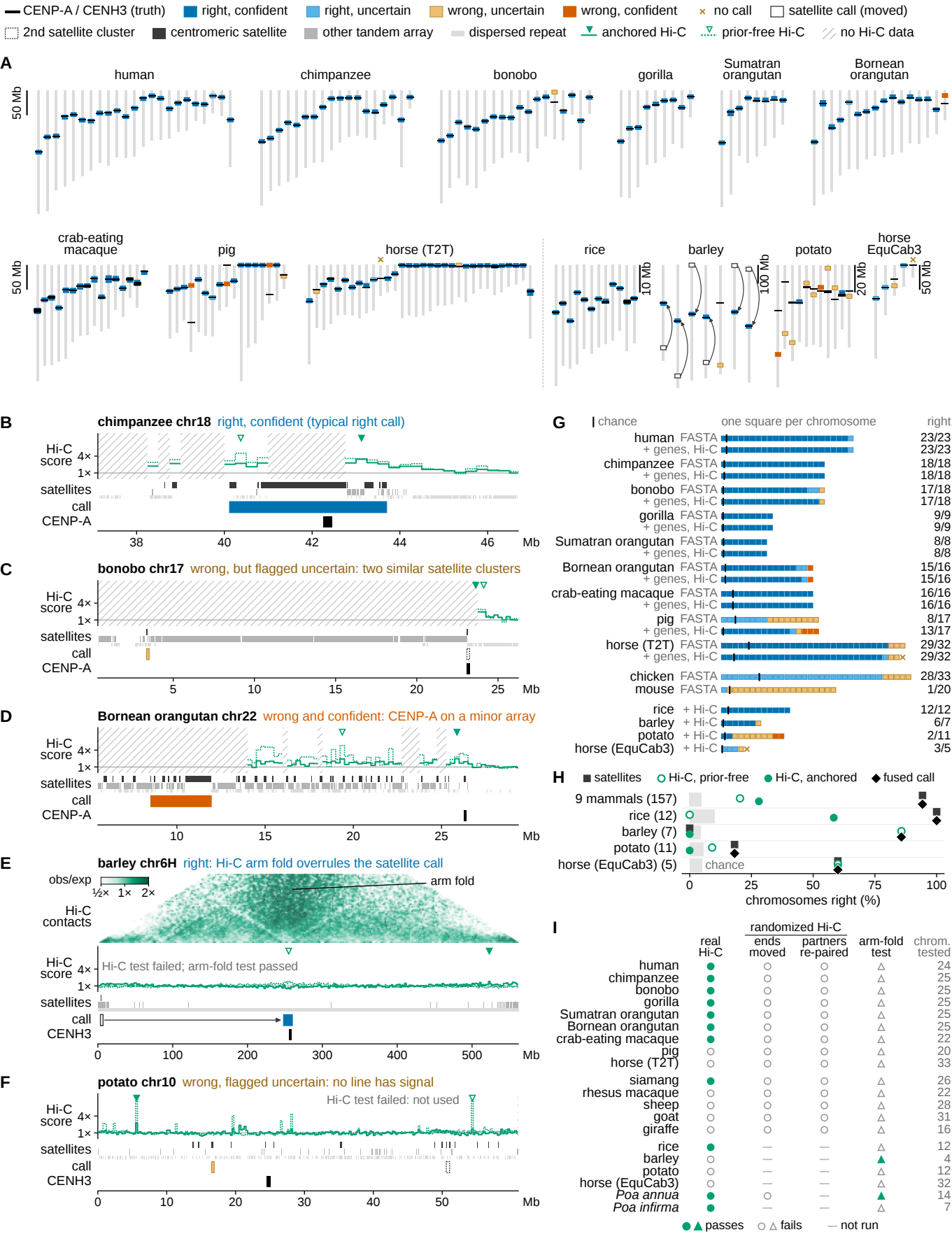

CENP-A / CENH3 (truth)
right, confident
right, uncertain
wrong, uncertain
wrong, confident
no call
satellite call (moved)
2nd satellite cluster
centromeric satellite
other tandem array
dispersed repeat
anchored Hi-C
prior-free Hi-C
no Hi-C data
A
human
chimpanzee
bonobo
gorilla
Sumatran orangutan
Bornean orangutan
crab-eating macaque
pig
horse (T2T)
rice
barley
potato
horse EquCab3
50 Mb
10 Mb
100 Mb
20 Mb
B
chimpanzee chr18 right, confident (typical right call)
C
bonobo chr17 wrong, but flagged uncertain: two similar satellite clusters
D
Bornean orangutan chr22 wrong and confident: CENP-A on a minor array
E
barley chr6H right: Hi-C arm fold overrules the satellite call
obs/exp
arm fold
Hi-C contacts
Hi-C test failed; arm-fold test passed
F
potato chr10 wrong, flagged uncertain: no line has signal
Hi-C test failed: not used
Hi-C score
satellites
call
CENP-A
CENH3
Mb
G
chance
one square per chromosome
right
human FASTA 23/23
+ genes, Hi-C 23/23
chimpanzee FASTA 18/18
+ genes, Hi-C 18/18
bonobo FASTA 17/18
+ genes, Hi-C 17/18
gorilla FASTA 9/9
+ genes, Hi-C 9/9
Sumatran orangutan FASTA 8/8
+ genes, Hi-C 8/8
Bornean orangutan FASTA 15/16
+ genes, Hi-C 15/16
crab-eating macaque FASTA 16/16
+ genes, Hi-C 16/16
pig FASTA 8/17
+ genes, Hi-C 13/17
horse (T2T) FASTA 29/32
+ genes, Hi-C 29/32
chicken FASTA 28/33
mouse FASTA 1/20
rice + Hi-C 12/12
barley + Hi-C 6/7
potato + Hi-C 2/11
horse (EquCab3) + Hi-C 3/5
H
satellites
Hi-C, prior-free
Hi-C, anchored
fused call
9 mammals (157)
rice (12)
barley (7)
potato (11)
horse (EquCab3) (5)
chance
chromosomes right (%)
I
randomized Hi-C
real Hi-C
ends moved
partners re-paired
arm-fold test
chrom. tested
human 24
chimpanzee 25
bonobo 25
gorilla 25
Sumatran orangutan 25
Bornean orangutan 25
crab-eating macaque 22
pig 20
horse (T2T) 33
siamang 26
rhesus macaque 22
sheep 28
goat 31
giraffe 16
rice 12
barley 4
potato 12
horse (EquCab3) 32
Poa annua 14
Poa infirma 7
passes
fails
not run

Fusing satellite repeats with a Hi-C signal that must first prove itself places 148 of 157 mammal centromeres on CENP-A and rescues barley centromeres that satellites miss; randomized Hi-C never passes the test. Where evidence is thin, cenfuse says “uncertain” instead of guessing.

A Every chromosome with a mapped centromere, to scale: the call (colored box) against the measured CENP-A or CENH3 domain (black bar). In mammals and rice the two coincide; barley's calls move away from a misleading satellite; most potato centromeres are missed, and flagged.

B A typical right call: the call covers the centromeric satellite array that carries CENP-A.

C Two similar satellite clusters: cenfuse picks one, names the other as the alternative and says it is unsure; CENP-A sits on the alternative.

D The confident miss: CENP-A sits on a few small arrays far from the large array that was called.

E Barley's satellite marks distal chromosome regions, not centromeres. The arm fold of a Rabl nucleus, seen in Hi-C, moves the call onto CENH3.

F A potato centromere with nothing to follow: no satellite under CENH3 and no Hi-C signal; the call is wrong and flagged.

G Every chromosome as one square. Nearly all mammal centromeres are right from the DNA sequence alone, far above chance; adding genes and Hi-C mostly raises confidence (and restores pig). Chicken calls are right but uncertain; mouse calls land on the wrong satellite at the right end.

H Each evidence line alone. Satellites work in mammals and rice, Hi-C in barley, neither in potato; the fused call matches the best line in every genome.

I Hi-C acts only after it finds the other chromosomes' centromeres. It does in apes, crab-eating macaque, rice and both *Poa* species, and never on randomized Hi-C.

Terms. Truth: CENP-A domains (centromere dip regions in great apes; CENP-A ChIP-seq in crab-eating macaque, pig and horse T2T, for pig and horse from other animals) and CENH3 or CENP-A ChIP-seq domains (rice, barley, potato, horse EquCab3). Right: the call overlaps the truth (mammal satellite and final calls) or lies within four Hi-C bins of it (every plant line, and mammal Hi-C peaks, which sit beside the unmappable array; bins 250 kb, human 500 kb, barley 2 Mb). Confident: confidence high or medium; uncertain: low. Chance: the probability that a call of the same length placed at random on the chromosome is right, summed over chromosomes. Runs: the v8 from-scratch rerun (*Poa* rows of I: see I). Every call is shown: “uncertain” is a designed output of the tool, and misses are shown on purpose.

A Gray bar: a chromosome hanging from its start; mammals and horse EquCab3 share one scale, rice, barley and potato have their own (scale bars). Box: final call with every input (mammals: FASTA, genes, Hi-C; plants: FASTA, Hi-C), colored by outcome, at least 0.75 mm tall; black bar: truth, at least 0.26 mm. Open boxes and arrows: barley's 6 satellite calls moved by Hi-C. Mammals 148/157 right (4 confidently wrong, 4 uncertain wrong, 1 no call; chance 7.6; $P = 1.9 \times 10^{-210}$, Poisson-binomial). Rice 12/12, barley 6/7, potato 2/11 (2 confidently wrong, 7 uncertain), horse EquCab3 3/5 (all uncertain); together 23/35, chance 2.3 ($P = 2.5 \times 10^{-20}$). Confident calls were right in 164 of 170 cases, uncertain or missing ones in 7 of 22 (Fisher's exact $P = 9.5 \times 10^{-13}$).

B–F Tracks: Hi-C score per bin, one scale for B–F (solid: anchored on the other chromosomes' satellite calls; dotted: prior-free; 1× = chromosome median; triangles: each line's peak; hatched: no Hi-C data, usually the unmappable array); satellite arrays by the role cenfuse gave them (dark: the family ranked centromeric; mid gray: other tandem arrays; light, thin: dispersed repeats, interspersed copies rejected as satellites; at least 0.12 mm wide); final call; truth. Windows: calls and truth ± 3 Mb (mammals), whole chromosome (plants). **B** was chosen by rule: of the great-ape right calls, the one nearest the medians of call width (3.3 Mb) and call-to-truth offset (0.45 Mb) of all 148 right mammal calls. **C** and **D** are the only great-ape misses. C: two small satellite clusters (22 and 23 kb); cenfuse took the first and reported the second, which lies under CENP-A and beside which both Hi-C peaks sit. D: CENP-A sits on 4 small arrays spanning 84 kb (73 kb of array) of families ranked centromeric, 14 Mb from the 3.4-Mb main array cluster that was called; only the anchored Hi-C peak points there, too weakly to overrule. **E**: the barley move of median length (lower median of 6); barley's satellite line follows a repeat family of distal chromosome regions, so the Hi-C test, anchored on those calls, fails; the arm-fold test passes and the call moves to the fold. Contact map: 836,986 cis read pairs, 4-Mb bins, balanced, observed/expected per distance, each pair of loci drawn above its midpoint, up to 300 Mb apart; in a Rabl nucleus the two arms lie side by side, so the fold appears as a ridge above the centromere. **F**: the median-distance miss among potato's 7 uncertain misses; small arrays of the centromeric family are scattered along the chromosome, none under CENH3, and Hi-C failed its test.

G One square per chromosome, sorted by outcome; black tick: chance expectation; numbers: right/total. Mammals, FASTA only: 143/157 right (13 of them uncertain), 13 wrong and uncertain, 0 without a call, 1 confidently wrong; chance 12.2. Every input: 148/157 right (4 uncertain), 4 wrong and uncertain, 1 without a call, 4 confidently wrong; chance 7.6. Pig from the FASTA alone is 8/17, all uncertain: in this rerun its satellite families split differently, no family dominates, and every call is capped at uncertain; with genes it is 13/17 and with genes and Hi-C 13/17 (3 of the 4 confident mammal misses; pig truth is ChIP-seq from other animals). Chicken (truth: CENP-A windows on 33 of 40 chromosomes): 28/33 right, all uncertain. Mouse (truth: the minor-satellite array; all chromosomes acrocentric): 1/20, all uncertain; every call sits at the centromeric end on the larger pericentric satellite, within 50 kb of the minor-satellite array on 20/20, which is the reported alternative on 20/20.

H Share of chromosomes right for each line alone (satellite call, prior-free Hi-C peak, anchored Hi-C peak) and for the fused call, every input; gray band: chance for the most generous line. Mammals: satellites 148, anchored Hi-C 44, prior-free Hi-C 32, fused 148 of 157. Rice: satellites 12/12. Barley: prior-free Hi-C 6/7, satellites 0/7. Potato: no line above 2/11. Plants pooled: satellites 17/35, prior-free Hi-C 10/35, anchored Hi-C 10/35, fused 23/35.

I Hi-C test, per chromosome: the other chromosomes' anchored Hi-C peaks must agree with their own confident calls on a majority, beyond chance (Poisson-binomial $P \le 0.01$). Within a genome it passed on every tested chromosome or on none. Real Hi-C passes in 8 of 14 mammals and in rice, *Poa annua* and *Poa infirma*; it fails in pig, horse T2T, rhesus macaque, sheep, goat, giraffe, barley, potato and horse EquCab3, where Hi-C then only reports (horse EquCab3 has no confident call, so its uncertain calls serve as references at clean peaks). Randomized Hi-C (read ends moved, or partners re-paired) passes on 0 of 708 chromosome tests in 29 runs and moves or places 0 calls. The arm-fold test (Rabl nuclei) passes only in barley and *Poa annua*, never on randomized Hi-C. Mammals without functional truth appear only here. *Poa* rows: fresh runs with the released code (satellite search included).

Exceptions. Horse T2T is 29/32 from the FASTA alone and 29/32 with every input, but not on the same chromosomes: genes and Hi-C fix 2 uncertain calls (CM128058.1, CM128066.1) and lose 2 (CM128050.1, uncertain, 2.7 Mb from CENP-A; CM128070.1, uncertain, 2.6 Mb from CENP-A); chr11 (CM128059.1, a satellite-free centromere) gets no call with every input (5.6 Mb off from the FASTA alone). Horse EquCab3 makes no confident call. The README's 33 randomized runs add 4 earlier runs on genomes whose calls are all uncertain (siamang without genes, chicken), not drawn here. Examples: B, E and F by the stated median rules; C and D because they are the only great-ape misses, so the typical failure cannot be chosen among more.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/update/figures/benchmark_overview · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/update/figures/benchmark_overview/benchmark_overview.py · 2026-10-30